BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 013 · Noughts & Crosses · Part 1 — Sequence before loops

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. In a zero-indexed C# array, which element does array[2] access?

Q2. Name the three parts of a C# for loop header used to visit every array element.

Q3. In a C# 2D array organised as rows and columns, what does GetLength(0) return?

Q4. If a board has 3 rows and 3 columns, how many values does it contain?

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Recall the term — sequence

Recall the term — statement

Recall the term — output

Recall the term — 2D array

Recall the term — row

Recall the term — column

Recall the term — index

New learning

Start with sequence — Write the five lines in the left panel into a C# console program. Run it. There are no loops ; each statement executes in order.

Make one simple change — Change the middle square 5 to X. Which exact line do you edit? This works — but changing the text every time a player moves is awkward. — CODE:

```
Console.WriteLine("4 | X | 6");
```

Store the squares as data — Put the nine square values into a 2D array. The first index chooses the row; the second chooses the column. — CODE:

```
string[,] board = {
{ "1", "2", "3" },
{ "4", "5", "6" },
{ "7", "8", "9" }
};
```

Display the data directly — Write the first row explicitly using three array accesses. Then do the same for rows 1 and 2. — CODE:

```
Console.WriteLine(board[0,0] + " | " +
board[0,1] + " | " +
board[0,2]);
```

Worked example

Make the board display before improving it

1. Create a working output — Run these instructions in order. The result is a 3×3 numbered board. — CODE:

```
Console.WriteLine("1 | 2 | 3");
Console.WriteLine("--+---+--");
```

Console.WriteLine("4 | 5 | 6");
Console.WriteLine("--+---+--");
Console.WriteLine("7 | 8 | 9");

2. Change one square — Change the centre of the printed board to X. This shows the limitation of storing positions inside string literals. — CODE:
Console.WriteLine("4 | X | 6");

3. Move the values into a 2D array — CODE:
string[,] board = {
{ "1", "2", "3" },
{ "4", "5", "6" },
{ "7", "8", "9" }
}; — Square 5 is board[1,1] , not board[5] , because our board uses zero-based row and column indexes.

4. Print the same values without a loop — CODE:
Console.WriteLine(board[0,0] + " | " + board[0,1] + " | " + board[0,2]);
Console.WriteLine(board[1,0] + " | " + board[1,1] + " | " + board[1,2]);
Console.WriteLine(board[2,0] + " | " + board[2,1] + " | " + board[2,2]); — This code works. But how many times do the row-printing instructions repeat? What changes between lines? We'll tackle that in Lesson 014.

We Do · Practise with guidance

Display three rows

Without loops, write three C# statements that print 1 | 2 | 3, then 4 | 5 | 6, then 7 | 8 | 9.

Write or explain

Why it works

Use one Console.WriteLine() for each row.

```
Console.WriteLine("1 | 2 | 3");
Console.WriteLine("4 | 5 | 6");
Console.WriteLine("7 | 8 | 9");
```

Find the square

With zero-based indexes, what is the row/column index of the centre square (5)?

The centre square is row 1, column 1.

```
board[1,1]
```

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Change the stored data

After declaring a 3×3 string array, write one assignment that places X in the centre.

```
board[1,1] = "X";
```

No other cell changes.

Spot repeated code

Look at three separate row-print statements. What is identical, and what changes each row?

Each statement prints three column values joined by separators. Only the row index changes: 0 → 1 → 2. This is repetition worth simplifying next lesson.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Create a static board
In .NET Fiddle, write sequential Console.WriteLine() statements to display the nine squares. Do not use a loop.

Question 2

Make a move by editing text
Change the centre displayed square from 5 to X by modifying one of your existing Console.WriteLine() statements.

Question 3

Create a 2D array
Declare string[,] board containing strings "1" through "9". Assign board[1,1] = "X"; .

Question 4

Print the array without loops
Using three separate Console.WriteLine() statements and explicit board[row,col] accesses, display all three rows. Highlight the parts of the code that repeat.

Exit Ticket · Answer before checking

We must write a loop before we can display a 3×3 board in C#.

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The item at index 2 — the third item.

Q2. A counter, condition and update.

Q3. The number of rows.

Q4. 9.

You Do pages · Independent questions

Q1. 1. Display the board with sequence
Each Console.WriteLine() runs in order. You can produce the whole board without a loop.
CODE:

```
Console.WriteLine("1 | 2 | 3");
Console.WriteLine("--+---+--");
Console.WriteLine("4 | 5 | 6");
Console.WriteLine("--+---+--");
Console.WriteLine("7 | 8 | 9");
```

Q2. 2. Change the middle value
Change the third WriteLine statement to show X rather than 5.
CODE:

```
Console.WriteLine("4 | X | 6");
```

Works, but makes moves difficult to maintain because the board is stored as printed text.

Q3. 3. Store the nine values in a two-dimensional array
Declare the board as string[,] and change its middle cell using row 1 and column 1.
CODE:

```
string[,] board = {
{ "1", "2", "3" },
{ "4", "5", "6" },
{ "7", "8", "9" }
};
board[1,1] = "X";
```

Indexes start at zero.

Q4. 4. Output data from the array without loops
The three statements still follow sequence, but they now display the array values. They work after a move changes an array element.
CODE:

```
Console.WriteLine(board[0,0] + " | " + board[0,1] + " | " + board[0,2]);
Console.WriteLine(board[1,0] + " | " + board[1,1] + " | " + board[1,2]);
Console.WriteLine(board[2,0] + " | " + board[2,1] + " | " + board[2,2]);
```

Notice: the only changing part of these statements is the row index. This is why iteration is useful next lesson.

Exit Ticket · Model answer

False. You can display a 3×3 board with explicit statements; a loop is useful but not required.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: